

DEEPAK MALLAMPATI

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Forward-deployed Applied AI engineer (2.5+ years) who ships agentic LLM workflows and RAG systems against messy real-world data - schema-validated agent contracts, RAG-grounded validation against domain documents, and audit-ready reasoning traces. Hands-on experience integrating with industrial enterprise systems (T-SQL stored procedures on a multi-billion-dollar oil-and-gas Order-to-Cash platform at Emerson) and shipping LLM apps end-to-end (Chrome extension with 4-provider multimodal payload handling on the Chrome Web Store, FastAPI/Docker production services). Comfortable embedding with users and turning operator workflows into working agents - quote intake, validation, duplicate detection, and risk-scoring patterns transfer from healthcare claims to industrial workflows.

CORE COMPETENCIES

Agentic LLM Workflows • Schema-Validated Agent Contracts • Forward-Deployed Engineering • RAG & Vector Retrieval • Multi-Step Tool Use • Customer-Embedded Delivery • Industrial ERP Integration (T-SQL, Synthesis O2C) • FastAPI/Flask + Docker Production • Multi-Provider LLM Payload Handling • Audit-Ready Reasoning Traces • Predictive ML (scikit-learn, XGBoost) • CI/CD (Jenkins) & Observability (Grafana)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare

- Shipped a 5-stage **agentic claims-processing pipeline** (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations; explainable risk scoring with audit-ready reasoning traces.
- Built **Retrieval-Augmented Generation (RAG)** systems over clinical knowledge and policy corpora; recursive semantic chunking + transformer embeddings improved contextual retrieval precision by **~35%**, reduced irrelevant retrieval by **>30%**; retrieval-grounded response alignment exceeded **90%** in evaluation.
- Developed **agentic LLM workflows** for multi-step queries (eligibility checks, workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by **~25%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by **~30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above **98%** and cut downstream model instability by **~40%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform (contracts, nominations, allocations, invoicing); reduced query execution time **~20%** and data retrieval latency **~25%** - direct industrial-ERP analog to mainframe/legacy back-ends.
- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL, and stored-procedure deployments; reduced deployment errors **>30%** and cycle time **~35-40%** via release validation and rollback checkpoints.
- Designed performance dashboards (SQL Server DMVs + Grafana) for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%** and improved root-cause resolution **~30%**.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing) and reconciliation queries; improved reconciliation runtime **~15%** and reduced deadlocks **~15%**.
- Supported RBAC and audit logging for financial modules in a compliance-sensitive oil & gas environment.

Machine Learning Engineer - Nonprofit Tech, Video Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; transcript preprocessing, hierarchical summarization, and timestamp-aligned clip extraction reduced manual review time by **60-70%**.
- Achieved **~85% alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs **~30%** and raised contextual answer accuracy **>85%**.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate-claim detection; explainable risk scoring with audit-ready reasoning traces - directly transferable to industrial quote/order/pricing/inventory agent contracts.

Manga Lens - Chrome Extension for Real-Time AI Translation shipped

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, `captureVisibleTab` for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline handles tall panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 sites, narrowed host permissions. Multi-provider payload handling (WebP for cloud, JPEG for Ollama).

Agentic Pixel Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis. Stable Diffusion fine-tuning + LoRA for identity, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

Dream Decoder - Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app interpreting dreams and synthesizing dreamscapes via coordinated multimodal APIs (Perplexity Sonar + Replicate). **Intermediate structured prompt transformation layers** improved contextual alignment **~30%** and first-pass image success **~25-30%** over naïve direct-prompt orchestration.

EDUCATION

Master's Degree - Kent State University

Aug 2023 - May 2025

Ohio, USA

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

Agentic AI & LLMs: Multi-agent orchestration, schema-validated agent contracts, tool discipline, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, RAG, semantic retrieval, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Industrial workflow automation (ERP/CRM/email integration), Healthcare AI (HIPAA-conscious), Applied AI, Forward-Deployed Engineering, Computer Vision, Multimodal Systems